Remote Sensing Semantic Segmentation via Boundary Supervision-Aided Multiscale Channelwise Cross Attention Network

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***Abstract*— High spatial resolution (HSR) remote sensing (RS) images inevitably pose the challenge of multiscale transformation, as small objects, such as cars and helicopters (HCs), may occupy only a few pixel points. This incurs a significant hurdle for global context modeling, particularly in backbone networks with large downsampling coefficients. Simple summation or concatenation techniques, such as skip connections, fail to address semantic gaps and even impose negative impacts on multiscale feature fusion. Meanwhile, due to the complexity of foreground objects, the boundary details of HSR RS images are easy to lose in sampling operations. To overcome these challenges, we propose a multiscale channelwise cross attention network (MCCANet) assisted by boundary supervision (BS). Technically, MCCA captures the channel attention (CA) with various scales, which allows dynamic and adaptive feature fusion in a contextual scale-aware manner and focuses on both large and small objects distributed throughout the inputs. Besides, a channel and context strainer (CCS) module is proposed and embedded in MCCA, filtering channels and contexts for the mitigation of intraclass differences. In addition, we apply a BS module to recover boundary contour, avoiding the blurring effect during the construction of contextual information. The refined boundary allows for the effective recognition of surrounding pixels, ensuring a better segmentation performance. Extensive experiments on the instance segmentation in aerial images dataset (iSAID), International Society for Photogrammetry and Remote Sensing (ISPRS) Potsdam, and land-cover domain adaptive (LoveDA) datasets demonstrate that our proposed MCCANet achieves a good balance of high accuracy and efficiency. Code will be available at: https://github.com/ZhengJianwei2/MCCANet.**

***Index Terms*— Attention module, boundary supervision (BS), convolutional neural network (CNN), remote sensing (RS), semantic segmentation.**

I. Introduction

HIGH spatial resolution (HSR) remote sensing (RS) images have been frequently emerged by the maturation of Earth observation technology. From these images, the objects of interest can be automatically extracted using approaches, such as semantic segmentation, which pixel-wisely assigns class labels and aids in road prospecting [1], urban planning [2], and land detection [3]. Compared with natural scenes, the semantic segmentation of HSR RS images poses specific challenges owing to the distinct imaging mechanisms and scene characteristics. In general, the practical challenges can be manifested in three aspects, as illustrated in Fig. 1.

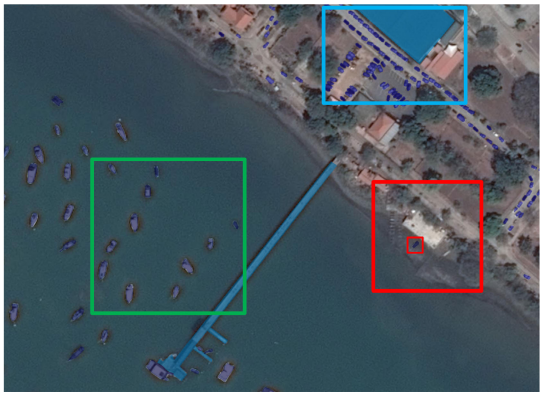


Fig. 1. Serious challenges in RS semantic segmentation. First, intensive intraclass differences are shown in green box. The shape and size of the SHs are different. Second, foreground–background imbalance is shown in red box. A vehicle is merged with the complex surroundings. Third, large-scale variations and complex boundaries are shown in blue box. The sizes of soccer field and vehicles hold obvious difference. Meanwhile, the boundaries of densely packed vehicles are ambiguous.

1) *Intensive Intraclass Differences:* Even objects in the same category differ greatly in their characteristics. More seriously, the foreground objects in HSR RS images are typically small, which occupy only a few pixel points and often lack sufficient information to enable accurate identification.
2) *Foreground–Background Imbalance:* The foreground objects of HSR RS images are often hidden in large-scale complex backgrounds. The highly complex backgrounds would incur significant false positives (FPs). In other words, it is difficult to effectively extract intrinsic characteristics from HSR RS images. In some datasets, the proportion of background to the whole image may even exceed 90% [4].
3) *Large-Scale Variations and Complex Boundaries:* Leaving aside the complex background, large-scale variations among different categories of foreground objects frequently happen. Moreover, many small objects, such as vehicles and ships (SHs), tend to appear at high densities, making the boundaries blurred and posing more difficulties.

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In recent years, deep convolutional neural networks (DCNNs) have been extensively applied in the field of semantic segmentation due to their superior capabilities of feature extraction and object representation. Technically, aggregation of multiscale context is known as the most important goals. For instance, the DeepLab series of networks [5], [6], [7] utilize atrous spatial pyramid pooling (ASPP) with varying receptive fields to achieve multiscale feature extraction. Pyramid scene parsing network (PSPNet) [8] aggregates context from different regions through a pyramid pooling module (PPM), allowing the model to capture global information. However, when applied to the RS fields, these approaches tend to ignore the issue of foreground–background imbalance, resulting in unsatisfactory performance. Apart from this, the encoder–decoder structures of [9] and [10] have also been successfully used for semantic segmentation. However, a succession of downsampling operations is involved in these networks, which may lead to the loss of high-frequency details and infringe on the segmentation effect at boundaries.

In addition to the aggregation of multiscale features, attention learning is another research direction on semantic segmentation [11], [12], which hammers at capturing the long-range dependencies among heterogeneous inputs. Besides, through the addition of boundary loss and certain preliminary branches [13], [14], [15], the boundary ambiguity issue has also incurred intensive investigation. Although these methods were originally designed for natural image segmentation, they have become the cornerstone for the development of RS-oriented models.

For the extreme cases of RS images, such as a large number of small foreground objects, ubiquitous occlusions, and complex boundary textures, several specific approaches have recently been proposed. For example, feature-selection high-resolution network (FSHRNet) [16] enforces feature fusion adaptively with high-resolution inputs, which could naturally alleviate the problem caused by large downsampling factors. FactSeg [17] enhances the recognition of weak features in small objects by foreground activation (FA) module, which together with a mixed loss leads to improved performance. Boundary-sensitive network (BSNet) [18] facilitates the extraction of boundary information by dynamic hybrid gradient convolution (DHGC) and simultaneously restores clear contours using coordinate sensitive attention (CSA).

Yet, with the prosperity of the HSR RS field, most semantic segmentation methods tackle only partially the mentioned challenges. For instance, [17] focuses on capturing the small objects yet neglects the precise feature learning for large targets, and [19] prioritizes overly the semantic information, paying little concern to the restoration of boundary textures. To cope with all the three challenges and cover the shortage of the existing solutions, we propose a new network architecture for the semantic segmentation of HSR RS images. Within the proposal, two modules, including a multiscale channelwise cross attention (MCCA) and a boundary supervision (BS), are, respectively, elaborated to aggregate multiscale contextual features and extract the target boundary, striving for better localization and identification of foreground objects. During the model optimization phase, both the objects and boundaries are supervised, with a focus on the discrimination of small targets and the enhanced classification of pixel points around the boundary. The contributions of this work can be summarized as follows.

1) *Network Architecture:* An MCCA network (MCCANet) employs a feature pyramid structure for channelwisely multiscale feature fusion in an attention-based manner. The small foreground objects are highlighted without compromising the detection of large objects. Besides, a boundary-decoupled branch is enforced to alleviate foreground–background imbalance.
2) *MCCA Module:* Technically, the proposed MCCA simultaneously considers large and intensively distributed regions as well as small and dispersedly distributed objects. Functionally, the multiscale feature fusion of MCCA is dynamic, adaptive, and also with the ability of contextual scale awareness. Moreover, by adaptively fusing a sufficient number of channel features, a channel and context strainer (CCS) subassembly is embedded to achieve a unification of semantic patterns, eliminating semantic gaps even with the complex background and severe intraclass differences.
3) *BS Module:* Semantic segmentation is often plagued by using inaccurate edge information. Accordingly, we propose the BS module to explicitly model and supervise object boundaries, which decouples the original feature map into two parts, i.e., the main body and the boundary. The former aggregates the contextual information inside the objects and effectively highlights the foreground. The latter boosts the generation of clear and smooth boundaries under supervision. During the optimization process, additional emphasis is placed on the points surrounding the boundaries for enhanced classification.

The performance of MCCANet is evaluated on three large-scale HSR RS image datasets, i.e., instance segmentation in aerial images dataset (iSAID) [20], International Society for Photogrammetry and Remote Sensing (ISPRS) Potsdam, and land-cover domain adaptive (LoveDA) [21]. Extensive experiments and ablation studies are conducted under the same conditions to assess the accuracy and efficiency of MCCANet and other competing methods. The experimental results demonstrate that the MCCANet achieves state-of-the-art performance in RS semantic segmentation.

The remainder of this article is organized as follows. Section II provides an overview of the related work, while Section III presents the general architecture of MCCANet and its key components. In Section IV, we compare MCCANet with other state-of-the-art semantic segmentation networks and conduct ablation experiments to assess the effectiveness of each module, along with analysis and visualization of the results. Finally, in Section V, our conclusions and future work are given.

II. Related Work

A. Semantic Segmentation of RS Image

Semantic segmentation of HSR images has a unique significance in the field of RS and is currently receiving increasing attention from massive researchers. In a similar vein to natural scene images, the segmentation of RS images has been greatly advanced by the development of deep learning. Due to the

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success of fully convolutional network (FCN) in the field of semantic segmentation [9], several extensions [22], [23] further enlarged the receptive field to retain the more spatial information. However, FCN and its subsequent work inevitably lose details due to downsampling operations, which seriously affect the segmentation of small objects and boundaries. On that basis, [24] and [25] mitigated the loss of spatial resolution by combining feature maps of different scales through skip connections.

The extraction of multiscale features through the more powerful networks has led a new paradigm, yet the extreme problems mentioned in Section I have not been solved, especially the ones of intensive intraclass variation and foreground–background imbalance. Accordingly, foreground-aware relation network (FarSeg) [4] was developed, which mainly uses the foreground-scene relation module to suppress false alarms. The PointFlow module [26] further solved the problem of FPs and foreground-awareness imbalance through pointwise affinity learning. Besides, hybrid multiple attention network (HMANet) [27] captured the correlation of features comprehensively from spatial, channel, and category perspectives to focus on the problem of intraclass variation. Comparatively, in this article, MCCANet focuses on aggregating multiscale contextual information through channel dimensions to alleviate the problem of foreground–background imbalance and large-scale variation.

B. Attention Mechanisms

Attention mechanisms in deep learning mimic human vision and can globally favor the assignment of the most informative feature expressions while suppressing the less useful ones. Vaswani et al. [28] first proposed self-attention mechanism to map the global dependencies of inputs and apply it to machine translation. Since then, attention mechanisms have been widely used in the field of image recognition, natural language processing, and speech analysis. For examples, squeeze-and-excitation network (SENet) [29] proposed an efficient and lightweight gating mechanism to self-calibrate the feature map through channel-directed import to weigh beneficial channels while suppressing less important ones. Bottleneck attention module (BAM) [30] and convolutional block attention module (CBAM) [31] introduced spatial features in a similar way to the channel attention (CA), encoding spatial information through large kernel convolutions.

As known, the attention module has the ability to improve the network performance with only a small number of additional parameters. Specific for the RS-oriented semantic segmentation, most improved attention methods were proposed in recent years. For example, semantic segmentation network with spatial and channel attention (SCAttNet) [32] integrated both spatial attention and CA to adaptively refine the features of foreground objects. Multiattention network (MANet) [33] proposed a kernel attention mechanism that can integrate local feature maps with the corresponding global dependencies. Following a similar line, class-constraint coarse-to-fine attentional deep network (CCAttNet) [34] strengthened the constraint of category features, thus obtaining long-range contextual information. SSAttNet [35] introduced a pyramid attention pooling module, achieving multiscale attention for better feature refinement. In practice, SSAttNet alleviates well both the issues of large intraclass variance and small interclass variance. To relieve the significant gap between high-level and low-level features, local attention network (LANet) [36] utilized a patch attention module (PAM) to gather contextual information from local patches that contain explicit semantic references. As a result, the pervasive semantic confusion in RS datasets is significantly reduced. Similarly, multiscale deformable attention net (MDANet) [37] reconsidered spatial attention mechanism from a local perspective, which employs deformable convolution to redefine spatial correlation, enhancing the ability to model long-distance relationships and adapt to the structure of RS images. In addition, [38], [39], [40], and [41] also designed some variants of the attention mechanism to exploit the more powerful global contextual information. Based on the success and advantages of the attention mechanisms mentioned above, we propose the MCCA module, which aggregates multiscale contextual information for better feature fusion.

C. Boundary Detection

Boundary detection plays a crucial role in a wide variety of computer vision problems. Yet, initially, the boundary features obtained are often shallow information, in which details are inevitably interspersed with some background noise. As the research progresses, the collection of boundary features becomes more in-depth and more accurate. For example, Luo et al. [13] adopted the traditional boundary detection operator Sobel, which is used to extract the edge of saliency graph. Zhou et al. [15] explored saliency and contour maps of multiple cues to enhance the network performance.

The concerned methods generally employ loss functions to assist in the achievement of boundary information. Another way to obtain boundary features refers to an additional branch. Wang et al. [14] proposed a saliency detection module to explicitly learn the boundaries of foreground objects. By localization flow, Su et al. [42] proposed a novel ground boundary-aware network to predict the refined feature maps. For the semantic segmentation of RS images, the edge part has also been emphasized to enhance the practical performance. For instance, Cheng et al. [43] designed a multitask network for segmentation and edge detection, both of which provide complementary cues for the gradient learning of the whole model. To earn for better contextual integration, Chong et al. [44] proposed a boundary flow and refined the contours of small-scale objects. In dual branches, Niu et al. [45] designed a decoupled learning paradigm to learn foreground and boundary separately. In this work, to inherit the advantages and meanwhile relieve the concerned shortcomings, we also propose a BS mechanism.

III. Methodology

In this section, the overall architecture of the proposed MCCANet is first given. Then, we delve into the detailed modules of MCCA, CCS, and BS, highlighting their unique contributions to the network. Finally, we outline the loss functions used in the training process.

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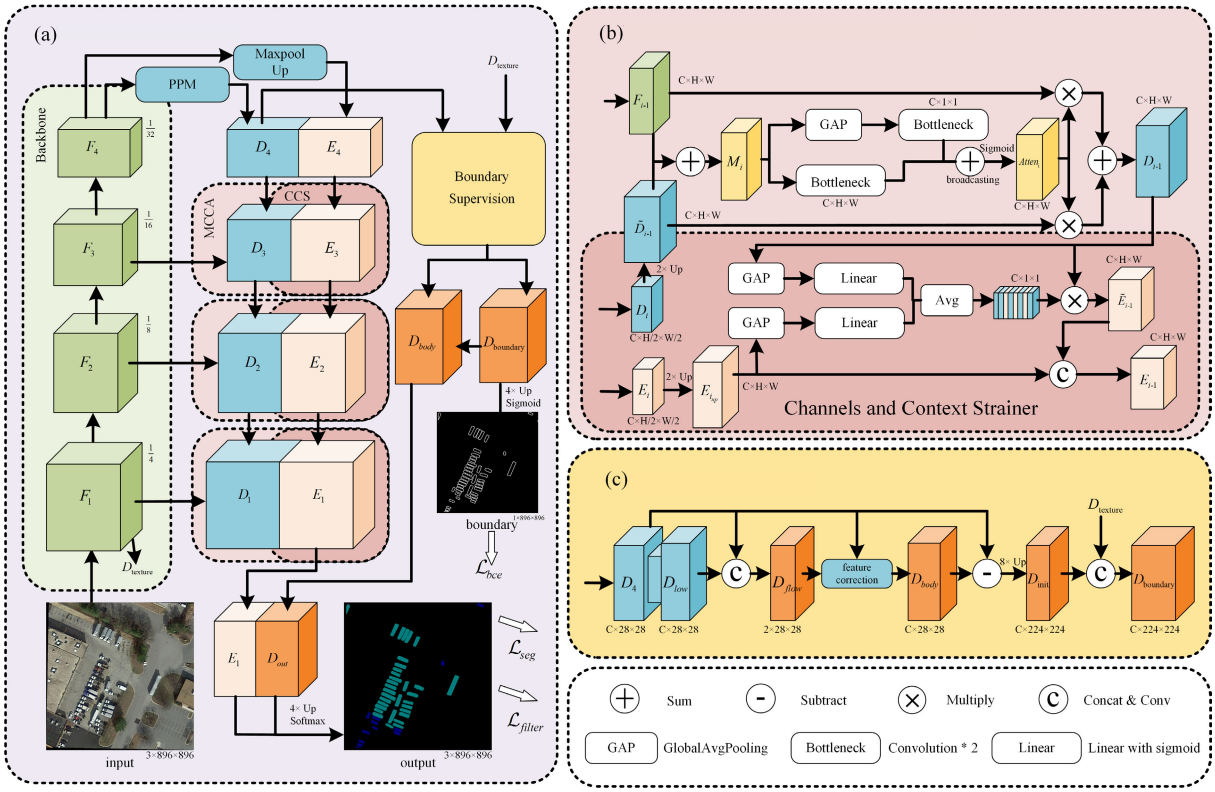


Fig. 2. Overview of the proposed MCCANet. (a) Network architecture. The green box indicates the backbone network, the pink box in the middle represents MCCA, and the yellow module on the right denotes BS. (b) MCCA module. (c) BS module.

A. Network Architecture

Fig. 2(a) roughly illustrates the architecture of MCCANet. Mainly, the network consists of a backbone of ResNet-50 [46] on the left, a three-layer MCCA module in the middle, and a BS module on the right. Functionally, the backbone network performs feature extraction in a bottom-up manner with four stages $\{F_1, F_2, F_3, F_4\}$. Based on the classical feature pyramid networks (FPNs), our MCCA module gradually fuses high-level semantic maps and low-level feature maps in three stages for multiscale information aggregation. Note that, in a contextual scale-aware manner, MCCA performs feature fusion dynamically and adaptively, which would prevent bias caused by aggregating contextual information on a global scale alone. In other words, the features concerning small objects would be highlighted, which is crucial for mitigating the issues of foreground–background imbalance and large-scale variation of foreground objects. Meanwhile, CCS guides the fused multiscale information to enforce an effective connection with decoder features, eliminating the semantic gap between the encoder and the decoder. Note that the achievement of unified semantic patterns is beneficial to tackling complex foreground objects of RS images, especially the ones within the same category. To further highlight the jagged foreground objects and outline the smooth boundaries, a module dubbed BS is provided, which decouples the original feature map into two parts: the main body and the boundary. The context across the foreground objects is aggregated to generate a clear main body, which relieves the foreground–background imbalance in depth. The boundary is clarified to enhance the visual effect of the segmentation result, which handles the dense occurrence of small targets effectively. By convolutional learning, the output features of MCCA and BS are finally connected to generate the segmentation map.

B. Multiscale Channelwise Cross Attention

Before giving the detailed description of MCCA, we first revisit CA that behaves as the prerequisites of our proposal. In a feature map, each channel represents a specialized detector, so that CA can focus on the more meaningful features. Among the applications, SENet [29] is the most compelling one, which squeezes each feature map of size $H \times W$ into a scalar. This extremely coarse descriptor prefers to emphasize large globally distributed objects and suffers the issue of erasing most of the useful signal present in small objects. Recall that there is a serious foreground–background imbalance problem in the segmentation of RS images, i.e., most objects are much smaller than the area of the background. Besides, large-scale variation among different categories of foreground objects is also a haunting issue. To make CA play an effective role in these cases, we propose to change the spatial pool and present a multiscale CA. On that basis, we also investigate feature fusion to aggregate contextual information. Thereupon, the overall MCCA module is proposed to integrate low-level features and high-level semantics.

To facilitate multiple-scale CA, the key idea lies in the change of spatial pool sizes. Unfortunately, too many sizes inevitably lead to a large number of parameters and a waste of computational resources. To keep a lightweight model,

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we choose to aggregate local and global contexts inside the CA. Local context is the first choice, as it is lightweight and can alleviate the bias of global contextual information. The aggregation of these two facilitates the better feature fusion weights Atten_i, which enable the module to dynamically fuse feature maps at different scales. The overall flowchart of MCCA is shown in Fig. 2(b).

With the operation of elementwise summation, we first generate the initial feature $M_i \in R^{C\times H\times W}$ from low-level feature map $F_{i-1} \in R^{C\times H\times W}$ and high-level feature map $D_i \in R^{C\times(H/2)\times(W/2)}$. Note PPM is employed to acquire the feature map D_4. Then, the multiscale CA weights $\text{Atten}_i \in R^{C\times H\times W}$ can be computed from the given M_i. Technically, two consecutive pointwise convolutions are employed to form a bottleneck structure $W(X)$ and aggregate the local channel context, in which a single pointwise convolution takes charge of achieving a channel relationship at each spatial position. The output of $W(X)$ holds the same shape as the input, which preserves the texture information within the low-level features. The global context is obtained similar to SENet [29], with the global average pooling (GAP) operation performed before the bottleneck structure. In practice, Atten_i is obtained by summing the local context $W(M_i) \in R^{C\times H\times W}$ and global context $W(\text{GAP}(M_i)) \in R^{C\times 1\times 1}$, followed by a sigmoid function, which can be further used as the weight to fuse low-level features and high-level semantics. The process is given as follows:

$$D_4 = \text{PPM}(F_4) \tag{1}$$

$$\widetilde{D}_{i-1} = \text{UpsamplingBilinear2d}(D_i) \tag{2}$$

$$M_i = F_{i-1} + \widetilde{D}_{i-1} \tag{3}$$

$$W(X) = \text{ReLU}(\text{BN}(\text{Conv2}(\text{ReLU}(\text{BN}(\text{Conv1}(X)))))) \tag{4}$$

$$\text{Atten}_i = \text{Sigmoid}(W(\text{GAP}(M_i)) + W(M_i)) \tag{5}$$

where $\widetilde{D}_{i-1}$ denotes the feature map upsampling on D_i, $M_i \in R^{C\times H\times W}$ represents the elementwise summation result of F_{i-1} and $\widetilde{D}_{i-1}$, $W(X)$ is the process of computing the channel relationship through the bottleneck structure, Conv1 and Conv2 are the pointwise convolutions for channel reduction and increase, respectively, BN is the batch normalization, ReLU is the rectified linear unit, $\text{GAP}(X) = 1/(H\times W)\sum_{j=1}^{H}\sum_{k=1}^{W} X_{i[:,j,k]}$ denotes the global average pooling, and sigmoid denotes the activation function.

Then, using Atten_i, we conduct a soft selection or weighted average between F_{i-1} and $\widetilde{D}_{i-1}$. In practice, the conduction borrows an elementwise multiplication to generate the fused feature map $D_{i-1} \in R^{C\times H\times W}$ for the next layer

$$D_{i-1} = \text{Atten}_i \times F_{i-1} + (1 - \text{Atten}_i) \times \widetilde{D}_{i-1} \tag{6}$$

where both Atten_i and $1-\text{Atten}_i$ are real numbers from 0 to 1 to achieve dynamic and adaptive feature fusion.

Merely fusing low-level features and high-level semantics by attention mechanism still suffer from certain inconsistencies. Therefore, with the aid of CCS assembly embedded in MCCA, we further introduce high-level features to strain channels and context and eliminate the semantic gap. CCS takes the fused feature map D_{i-1} and the high-level features $E_i \in R^{C\times(H/2)\times(W/2)}$ as inputs. D_{i-1} and the upsampled $E_{i_{\text{up}}} \in R^{C\times H\times W}$ are spatially squeezed through GAP, embedding global information and generating feature vectors. A linear layer with the same size of inputs and outputs is used to represent the channelwise dependencies. The above two operations are represented by $H(X)$. Then, by activating the average of $H(D_{i-1})$ and $H(E_{i_{\text{up}}})$ through the sigmoid function, an attention map representing the importance of the channel can be obtained to recalibrate D_{i-1}. The adjusted feature map $\widetilde{E}_{i-1} \in R^{C\times H\times W}$ is connected back to $E_{i_{\text{up}}}$ and then learned by convolution to achieve the unification of semantic patterns with sufficient channelwise features, obtaining the output $E_{i-1} \in R^{C\times H\times W}$ of the whole MCCA. The specific operations are as follows:

$$E_4 = \text{Up}(\text{Conv}(\text{Maxpool}(F_4))) \tag{7}$$

$$E_{i_{\text{up}}} = \text{UpsamplingBilinear2d}(E_i) \tag{8}$$

$$H(X) = \text{Linear}(\text{GAP}(X)) \tag{9}$$

$$\widetilde{E}_{i-1} = \text{Sigmoid}\big(\text{Avg}\big(H(D_{i-1}), H\big(E_{i_{\text{up}}}\big)\big)\big) \times D_{i-1} \tag{10}$$

$$E_{i-1} = \text{Conv}\Big(\text{Concat}\Big(E_{i_{\text{up}}}, \widetilde{E}_{i-1}\Big)\Big) \tag{11}$$

where $E_4 \in R^{C\times 28\times 28}$ is obtained from the last layer of the backbone network by downscaling (two times) with maxpool convolution and then upsampling to the same size of F_4. Afterward, E_4 is used as the input for the first computation of CCS module. $E_{i_{\text{up}}}$ is the upsampled E_i with the same size as D_{i-1}. $H(X)$ indicates GAP followed by a linear layer. Linear denotes a linear layer with the same input and output sizes.

C. Boundary Supervision

MCCA assists in locating and identifying small targets well, yet the detected objects lack a perfect boundary contour. To sketch a better profile, we prob into the boundary feature map of the detected objects as a supervision. Surprisingly, during the generation of boundary features, the main body part favors the refinement of the original boundary map and highlights the foreground objects, especially those hidden in weak positions. The structure of the BS module is illustrated in Fig. 2(c), whose main idea lies in the enrichment of the main body features, followed by a subtraction operation for the obtainment of boundary maps.

Flow field generation and feature correction are employed to generate the main body part. Concretely, we downsample the original feature map $D_4 \in R^{C\times 28\times 28}$ to a low-resolution form, giving it a low spatial frequency. The reduction of resolution helps to find a set holding the central or seed points, from which the higher-level semantic features can be highlighted. Note that given F_4, the original feature map D_4 is generated by PPM due to its superior effectiveness in capturing contextual information. Then, it is upsampled to the same size as the original feature map, obtaining $D_{\text{low}} \in R^{C\times 28\times 28}$. Downsampling is carried by two successive 3×3 convolutions of stride 2. Besides, upsampling is realized by bilinear interpolation operation. With D_4 and D_{low} concatenated, we use another 3×3 convolution to predict the flow field map $D_{\text{flow}} \in R^{2\times 28\times 28}$. The process of flow field generation can

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be formulated as follows:

$$D_{\text{low}} = \text{Up}(\text{CBR}(\text{CBR}(D_4))) \tag{12}$$

$$D_{\text{flow}} = \text{Conv}(\text{Concat}(D_4, D_{\text{low}})) \tag{13}$$

where convolution + batchnorm + ReLU (CBR) denotes the process of 3×3 convolution, BatchNorm2d, and ReLU.

With the aid of D_{flow}, the original feature map D_4 can be corrected through flow field maps. Specifically, a differentiable bilinear sampling mechanism similar to that in spatial transformer networks [47] approximates each point to obtain the main body map $D_{\text{body}} \in R^{C\times 28\times 28}$. The process is given in the following equation:

$$D_{\text{body}}(p_x) = \sum_{p\in\mathcal{N}(p_i)} \mathcal{W}(D_{\text{flow}}) D_4(p) \tag{14}$$

where p_i is an arbitrary point in space, $\mathcal{N}$ represents the set of points around p_i, $\mathcal{W}$ denotes the bilinear kernel weights on warped spatial grids, and p_x is a point in the computed D_{body}.

Note the boundary D_{boundary} holds the more high-frequency components relative to the main body D_{body}. Given this assumption, we subtract D_{body} from the original feature map D_4 and upsample it to obtain $D_{\text{init}} \in R^{C\times 224\times 224}$, followed by the addition of low-level feature $D_{\text{texture}} \in R^{C\times 224\times 224}$, which would enhance the high-frequency terms and supplement the missed details. Moreover, $D_{\text{boundary}} \in R^{C\times 224\times 224}$ is achieved by using 1×1 convolution. The final boundary $\in R^{1\times 896\times 896}$ is then generated by convolving and upsampling D_{boundary} and activated by the sigmoid function

$$D_{\text{init}} = \text{Up8*}\left(D_4 - D_{\text{body}}\right) \tag{15}$$

$$D_{\text{boundary}} = \text{Conv}(\text{Concat}(D_{\text{init}}, D_{\text{texture}})) \tag{16}$$

$$\text{boundary} = \text{Sigmoid}\left(\text{Up4*}\left(\text{Conv}\left(D_{\text{boundary}}\right)\right)\right) \tag{17}$$

where Up8* and Up4* denote $8\times$ and $4\times$ upsampling, respectively. D_{texture} is obtained from the first layer F_1 of the backbone by 1×1 convolutional operation.

To maximize the respective roles of D_{body} and D_{boundary}, we merge the two and generate a refined feature map $D_{\text{out}} \in R^{2C\times 224\times 224}$ to complement the output E_1 of MCCA, enhancing the foreground and boundary saliency of the overall output. D_{out} is first generated by concatenating D_4 with the sum of D_{body} and D_{boundary}. Afterward, by combining D_{out} and E_1, we achieve the final feature map Output $\in R^{K\times 896\times 896}$ through convolutional learning as well as upsampling and a softmax classifier. K represents the number of categories. The process can be expressed as follows:

$$D_{\text{out}} = \text{Concat}\left(\text{Up8*}(D_4), \text{Up8*}\left(D_{\text{body}}\right) + D_{\text{boundary}}\right) \tag{18}$$

$$\text{Output} = \text{Softmax}(\text{Up4*}(\text{CBR3}(\text{Concat}(D_{\text{out}}, E_1)))) \tag{19}$$

where D_{out} can be considered as the final output of the BS module. CBR3 indicates the process of two consecutive 3×3 convolutions and one 1×1 convolution. Softmax indicates the category classifier.

In order to illustrate the process of obtaining segmentation results more intuitively, we visualize the steps briefly in Fig. 3. Fig. 3(b) is the flow field learned from Fig. 3(a). It is worthy to note that the flow field is somewhat blurred, because the generation process is performed in a high semantic environment with low resolution. Fig. 3(c) shows the distinct boundary part by subtracting the main body of the generated flow field. Fig. 3(d) shows the final feature map of the whole network, which possesses richer expressiveness in both the main body part and the boundary.

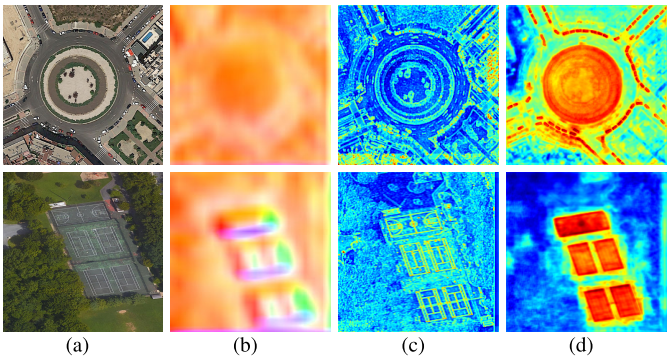


Fig. 3. Visualization of feature maps in several steps. (a) Original images. (b) Learned flow field. (c) Feature maps of boundaries. (d) Final feature maps.

D. Loss Function

For the supervision of the final results, we choose cross entropy (CE) loss with unpredictable example estimation capability

$$w = \frac{1}{Z}(1-p)^{\gamma} \tag{20}$$

$$\mathcal{L}_{\text{seg}}(p, y) = [w + \zeta(t)(1-w)] \cdot \mathcal{L}_{\text{ce}}(p, y) \tag{21}$$

where p and y denote the predicted result and ground truth (GT), respectively, and w is used to denote the unpredictable example weight. The operation of $1-p$ can invert the predicted probability of our result $p \in [0, 1]$ as focal loss, making the network focus more on the unpredictable example. Z and γ are two constants to correct the weights. $\zeta(t)$ denotes an annealing function $0.5*(1+\cos(t/\text{annealing_step}\cdot\pi))$ with respect to the current training step t. $\mathcal{L}_{\text{ce}}(p, y)$ denotes the multiclass CE loss.

In addition to supervision for the final output, we also deliberately introduce auxiliary supervision losses for the boundary subassembly. The first is the binary classification of the boundaries, i.e., to determine whether the prediction of the boundary is correct or not, in which case the CE loss degrades into a binary CE (BCE) loss. Apart from that, a comprehensive loss based on the boundary prior judgment is introduced, which mainly tackles the difficulty of classifying the points around the boundary. Specifically, with the boundary predicted, the edge points and their surroundings are used to update the traditional CE loss. In other words, our CE loss is performed for those points satisfying the following two conditions.

1) The points locate precisely at the boundary.
2) The points around the boundary, which are mainly wrongly detected or missed.

The conditions are judged by the preset thresholds. Overall, the auxiliary loss of boundary can be formulated

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as follows:

$$\mathcal{L}_{\text{bce}} = -\left[y_b \log(p_b) + (1 - y_b)\log(1 - p_b)\right] \tag{22}$$

$$\mathcal{L}_{\text{filter}} = -\frac{1}{K}\sum_{i=1}^{N}\mathbb{I}\left(p_j < \alpha\right) \cdot \log p_j, \quad p_j \in \mathcal{N}(p_i) \tag{23}$$

where p_b and y_b are the predicted result and GT of the boundary. In (23), N is the sum of all pixels, K is set to one-tenth of N, and p_j denotes a point in the domain $\mathcal{N}$ of the boundary point p_i. The domain $\mathcal{N}$ is predefined for those points with confidence less than α. $\mathbb{I}$ is an indicator factor, which equals 1 when the condition is satisfied and 0 otherwise.

Combining these losses, the total loss $\mathcal{L}$ can be expressed as follows:

$$\mathcal{L} = \lambda_1\mathcal{L}_{\text{seg}} + \lambda_2\mathcal{L}_{\text{bce}} + \lambda_3\mathcal{L}_{\text{filter}} \tag{24}$$

where λ_1–λ_3 are three hyperparameters used to control each loss, which can be adjusted depending on the characteristics of the dataset. In our experiments, we have tried various numerical ratios of the three hyperparameters. Relatively speaking, the best results are achieved when the ratios fixed at λ_1:λ_2:λ_3 = 1:2:1.

IV. Experiments

A. Datasets Description

In order to verify the performance of the proposed MCCANet, we conduct extensive experiments on three representative aerial imagery datasets.

1) iSAID: iSAID is currently the largest RS dataset with a foreground–background imbalance problem. The dataset contains 655 451 instances with 15 categories on 2806 HSR optical images, whose size ranges from 12 029 × 5014 to 455 × 387. We use 1411 images for training and 458 images for evaluation of the proposed method. Note the semantic annotations for the 937 images in the test set are not available.

2) Potsdam: ISPRS Potsdam contains 38 HSR images of size 6000 × 6000. These images cover a large area and are highly variable. Each image contains true orthophoto (TOP) extracted from the larger TOP mosaic. The ground sampling distance of TOP is 5 cm. The dataset provides three different channels composing the data. In this work, we use red green blue (RGB) data for training and evaluation. Because the image of 7_10 suffers error annotations [59], we select 23 images to train the model and use the remaining 14 images to test the performance of the model. The test set contains images of 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, 4_15, 5_13, 5_14, 5_15, 6_13, 6_14, 6_15, and 7_13.

3) LoveDA: The LoveDA dataset contains 5987 HSR images and 166 768 labeled objects from three different cities in China (Nanjing, Changzhou, and Wuhan). Different geographical landscapes of urban and rural scenes can be visualized. The images are of 1024 × 1024 pixels and include seven land-cover categories, i.e., building, road, water, barren, forest, agriculture, and background. The 2522 images are used for training, 1669 images are used for validation, and 1796 images are officially provided for testing. The challenges of this dataset are known as multiscale objects, complex background samples, and inconsistent class distributions.

B. Implementation Details

1) Training Settings: In the training process, a stochastic gradient descent (SGD) optimizer is used with a momentum of 0.9 and a weight decay of 0.0001. The initial learning rate is set to 0.007. Besides, the learning rate is dynamically adjusted during the training process using a poly strategy, which can be expressed as follows:

$$\text{lr} = \text{base_lr} \times \left(1 - \frac{\text{epoch}}{\text{num_epoch}}\right)^{\text{power}} \tag{25}$$

where lr is the new learning rate, base_lr is the baseline learning rate, epoch is the number of iterations, num_epoch is the maximum number of iterations, and power is set to 0.9 in the experiment to control the shape of the curve.

On iSAID, we use a sliding window whose stride is 512 to randomly crop patches of size 896 × 896 from the original images. Then, the patches are mirrored and rotated randomly, and the batch size is set to 4 per GPU. The experiments are conducted on two NVIDIA RTX 3090 GPUs for 16 epochs using the PyTorch framework. Experiments on the Potsdam and LoveDA datasets differed slightly from the iSAID dataset by cropping the patches to a smaller size of 512 × 512. Meanwhile, the initial learning rates are adjusted to 0.03 for Potsdam and 0.01 for LoveDA, respectively. Due to the relatively small size of these two datasets, 200 epochs of training are conducted.

2) Evaluation Metrics: We mainly use intersection over union (IoU) and $F1$-score ($F1$) to evaluate the performance of the network. Each metric is individually defined as follows:

$$\text{IoU} = \text{TP}/(\text{TP} + \text{FP} + \text{FN}) \tag{26}$$

$$\text{Precision} = \text{TP}/(\text{TP} + \text{FP}) \tag{27}$$

$$\text{Recall} = \text{TP}/(\text{TP} + \text{FN}) \tag{28}$$

$$F1 = 2/\left(\text{Recall}^{-1} + \text{Precision}^{-1}\right) \tag{29}$$

where TP, TN, FP, and FN represent the number of true positive, true negative, false positive, and false negative, respectively. Notably, $F1$ is a composite metric that integrates both precision and recall.

C. Comparison Experiments

1) Comparisons on the iSAID Dataset: For a comprehensive comparison, massive state-of-the-art methods are selected as the competitors, such as dual attention network (DANet) [12], DeepLab v3+ [7], Semantic FPN [48], PSPNet [8], DenseASPP [49], segmentation transformer (SETR) [50], UperNet [51], AlignSeg [52], HMANet [27], BSNet [18], FarSeg [4], FactSeg [17], RS pretraining (RSP) [53], UNetFormer [54], rotated varied-size attention (RVSA) [55], graph reasoning and disentangled learning (GR&DL) [45], and RS segmentation framework with transformer (RSSFormer) [19]. All the experiments are conducted under the same conditions. Note that most of the networks use ResNet-50 as the backbone, while a few models use transformer-based backbones, such as T-Large and Swin-B. Due to the advancement in computing power, it has become feasible and increasingly popular to pretrain backbone networks through RS images. Therefore,

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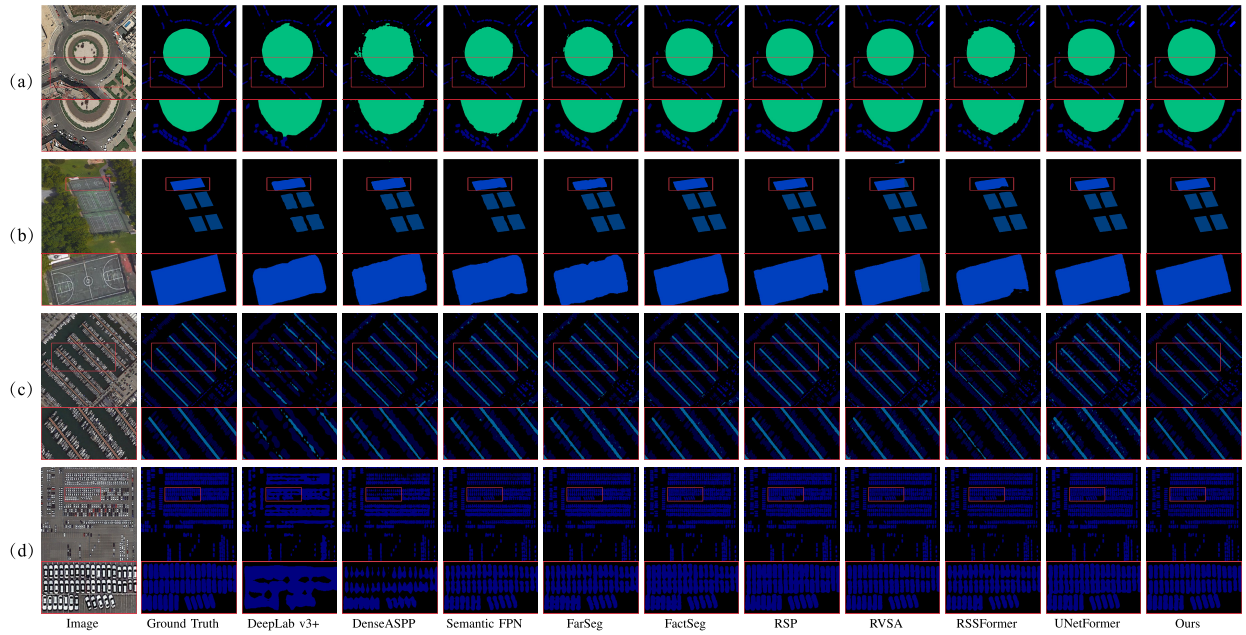


Fig. 4. Visualization results on the iSAID validation set. (a) Storage tank and small vehicle, (b) basketball court and tennis court, (c) harbor and ship, and (d) small vehicle.

TABLE I

COMPARISONS WITH STATE OF THE ARTS ON iSAID VALIDATION SET. THE MEANING OF THE ABBREVIATIONS OF CATEGORIES IS AS FOLLOWS. PL: PLANE. BD: BASEBALL DIAMOND. BR: BRIDGE. GTF: GROUND TRACK FIELD. SV: SMALL VEHICLE. LV: LARGE VEHICLE. SH: SHIP. TC: TENNIS COURT. BC: BASKETBALL COURT. ST: STORAGE TANK. SBF: SOCCER BALL FIELD. RA: ROUNDABOUT. HA: HARBOR. SP: SWIMMING POOL. HC: HELICOPTER

Method	Backbone	IoU per category(%)															mIoU(%)	mean F1
		PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC		
DANet [12]	ResNet-50	80.9	71.4	28.6	52.5	42.1	57.5	60.2	84.7	50.9	63.0	63.3	40.6	48.8	46.1	30.4	57.5	71.0
DeepLab v3+ [7]	ResNet-50	76.2	75.9	34.3	58.2	33.6	54.5	59.9	84.2	58.6	58.0	74.0	66.0	44.8	45.8	30.5	59.6	72.9
Semantic FPN [48]	ResNet-50	79.0	73.3	31.5	48.3	42.6	56.0	62.1	84.2	58.3	56.9	71.7	66.9	50.4	44.1	33.6	59.7	73.1
PSPNet [8]	ResNet-50	82.5	76.8	31.2	53.4	47.3	58.2	63.9	86.5	56.7	68.3	67.4	52.6	53.0	44.0	33.3	60.9	74.2
DenseASPP [49]	DenseNet-121	80.2	76.4	37.9	55.7	46.3	57.5	64.5	86.2	62.4	54.5	61.2	69.4	53.2	47.3	24.9	61.0	74.3
SETR [50]	T-Large	81.5	74.5	30.8	52.3	48.8	58.6	63.4	86.9	57.4	67.4	68.0	61.6	52.9	46.2	32.6	61.4	74.4
UperNet [51]	Swin-B	83.0	75.3	30.5	51.8	49.6	59.2	63.1	85.7	59.2	68.2	68.1	59.5	53.3	46.8	34.3	61.6	74.7
AlignSeg [52]	ResNet-50	82.9	76.2	28.7	52.0	50.3	60.7	67.4	86.2	62.1	68.9	71.2	56.2	54.8	45.7	31.2	62.1	75.0
HMANet [27]	ResNet-50	83.8	74.7	29.0	54.6	50.3	59.7	65.4	88.7	60.5	70.9	70.2	62.9	51.9	51.4	32.6	62.6	75.5
BSNet [18]	ResNet-50	81.8	76.1	41.3	50.3	46.6	63.4	65.3	86.8	55.9	69.1	70.2	70.0	57.3	48.8	31.8	63.4	76.6
FarSeg [4]	ResNet-50	82.0	77.7	36.7	56.7	46.3	60.6	65.4	86.4	62.1	61.8	72.5	71.4	53.9	51.2	35.8	63.7	76.5
FactSeg [17]	ResNet-50	84.1	78.3	36.3	54.6	49.5	62.6	68.3	88.9	64.8	56.8	73.5	69.4	55.7	51.4	42.7	64.8	77.3
RSP [53]	ViTAEv2-S	85.1	72.2	44.0	**63.0**	51.6	62.5	71.3	70.4	57.3	74.3	**77.6**	62.2	54.7	46.9	35.7	64.3	77.1
UNetFormer [54]	ResNet-18	84.4	76.8	39.9	56.4	52.1	63.4	69.0	87.8	60.3	63.4	73.9	71.1	56.5	44.5	31.2	64.4	77.3
RVSA [55]	ViTAE-B	85.7	73.1	**44.7**	59.6	52.5	63.8	70.6	70.5	53.4	**77.1**	74.6	66.5	54.9	47.0	38.0	64.5	77.3
GR&DL [45]	ResNet-50	84.2	**78.5**	38.1	56.9	50.7	60.5	65.5	88.7	**63.5**	69.2	73.7	68.7	52.8	52.4	38.6	65.1	77.4
RSSFormer [19]	RSS-B	85.0	76.0	37.6	59.1	51.2	62.9	69.5	88.2	62.1	64.0	73.0	71.9	55.0	51.3	36.1	65.2	78.2
Ours	ResNet-50	**86.4**	77.0	38.9	59.9	**52.6**	**63.9**	**71.6**	**89.7**	63.4	65.9	74.6	**73.1**	**58.7**	**52.8**	**43.6**	**66.9**	**79.1**

the RS foundation models, such as RSP [53] and RVSA [55], have also been included in the comparison.

Fig. 4 and Table I give the qualitative and quantitative results, respectively. To save space, Fig. 4 selects some representative competitors, including mainstream DeepLab v3+, DenseASPP, Semantic FPN, FarSeg, FactSeg, RSP, RVSA, RSSFormer, and UNetFormer. In addition, to better show the segmentation results, we have performed a local zoom effect on the final recoveries. Besides, to illustrate the gaps between different models more visually, the qualitative comparison of prediction errors is also provided in Fig. 5. It can be observed intuitively that combined with the error plots in Fig. 5(a), the dual-branch decoder of FactSeg can obtain the activation and refinement features, yet it cannot effectively deal with the boundary. Fig. 4(b) demonstrates the accuracy of our model for the boundary judgment, in which only our segmentation results present the clear edges, while the ones of other models are all distorted. As shown in Fig. 5(d), both DeepLab v3+ and DenseASPP generate poor segmentation results for small targets, especially when they occur densely. For DeepLab v3+, the presence of large downsampling coefficients leads to sticking intervals. Accordingly, DenseASPP identifies the targets with fuzzy edges. More evidently, as given in Fig. 4(c), our model not only maintains the straightest dock lines,

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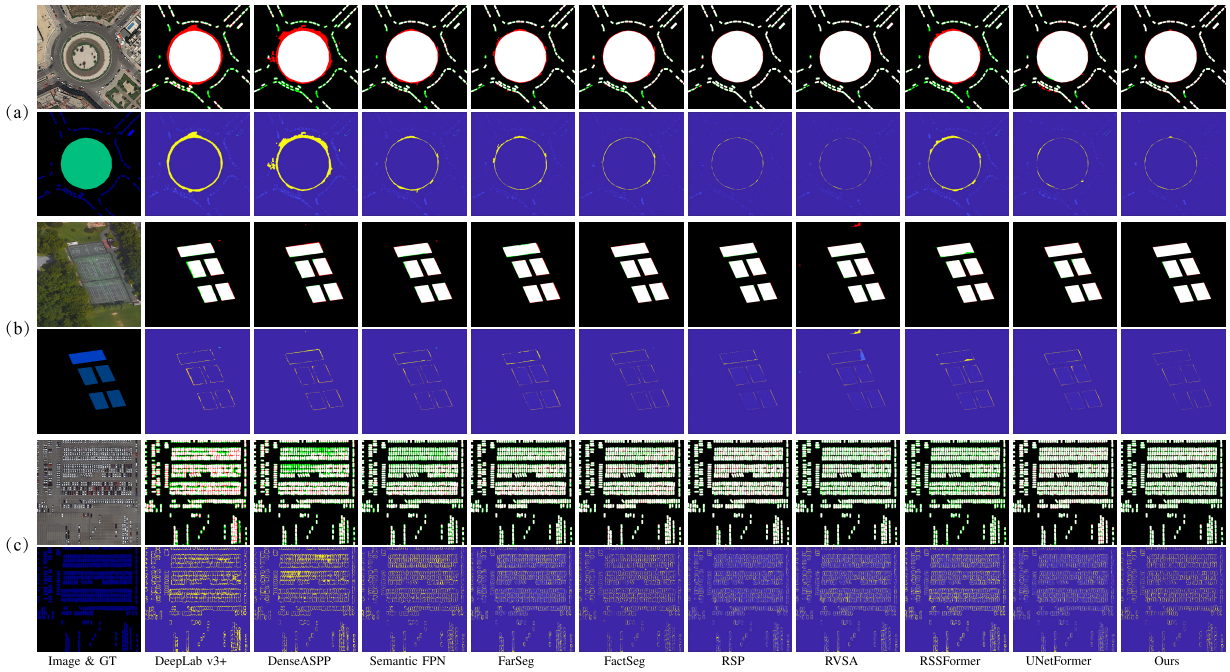


Fig. 5. Visual results of different models on iSAID, where the odd-numbered rows are the segmentation predictions and the even-numbered rows are the error maps. (a)–(c) Three representative samples. GT is the ground truth. Different colors are used to facilitate a better observation. In segmentation predictions, white is a true positive, black is a true negative, red indicates an FP, and green stands as a false negative. In error maps, yellow is error detection, and blue is the correct region.

TABLE II
COMPARISONS WITH STATE OF THE ARTS ON THE POTSDAM TEST SET

Method	Backbone	F1 per category(%)					mean F1	mIoU(%)
		ImpSurf	Building	LowVeg	Tree	Car		
FCN-8s [9]	VGG-16	87.3	91.8	81.2	83.0	87.1	86.1	76.3
ResUNet [56]	ResNet-50	88.3	92.9	82.6	84.0	88.6	87.2	78.2
Semantic FPN [48]	ResNet-50	89.3	94.1	83.2	84.4	88.5	87.9	79.3
DeepLab v3+ [7]	ResNet-50	89.9	94.7	83.8	85.5	88.5	88.4	79.6
PSPNet [8]	ResNet-50	90.1	95.0	83.9	84.9	89.3	88.6	80.1
DenseASPP [49]	DenseNet-121	90.2	95.1	84.1	85.4	89.8	88.9	80.2
FarSeg [4]	ResNet-50	90.5	**95.3**	83.9	85.1	90.4	89.0	80.4
FactSeg [17]	ResNet-50	90.4	95.1	84.4	85.5	90.9	89.2	80.5
BANet [57]	ResT-Lite	90.5	94.9	83.9	84.8	90.3	88.9	80.3
UNetFormer [54]	ResNet-18	90.3	95.0	84.2	85.6	90.5	89.1	80.6
RSSFormer [19]	RSS-B	90.2	95.3	84.0	84.6	88.7	88.6	79.8
Ours	ResNet-50	**90.5**	95.2	**84.8**	**85.9**	**91.1**	**89.5**	**81.2**

but also achieves the more impressive results in the segmentation of various types of SHs. Besides, for MCCANet, little error happens even in the presence of large intraclass differences.

Overall, both the visualizations in Figs. 4 and 5 show that our MCCANet achieves much better results. Table I also supports the superiority of our model, obtaining the highest mean IoU (mIoU) on the validation set of the iSAID dataset. Noteworthy, we achieve a more pronounced lead on targets, such as helicopters (HCs) and small vehicles (SVs), which are often much too small, occupying only a trivial proportion to the whole image.

2) Comparisons on the Potsdam Dataset: Unlike the iSAID dataset that focuses extraordinarily on the foreground–background imbalance problem, the Potsdam dataset is an aerial view of a city. The numerical results of each model on the test set are given in Table II. It can be clearly seen that our proposed method achieves excellent results, with the best mean $F1$ score of 89.5 and the best mIoU score of 81.2. In terms of different categories, MCCANet also dominates most of the best values, especially for the cases of car and low vegetation in which a large number of small targets exist.

The qualitative results from FCN-8s [9], DeepLab v3+ [7], Semantic FPN [48], DenseASPP [49], FarSeg [4], FactSeg [17], RSSFormer [19], UNetFormer [54], BANet [57], and our MCCANet are visualized in Fig. 6. Consistent with the numerical results, the visualization shows that our method predicts more accurately in different foreground objects, especially the ones with small size. The learned boundaries of our method are also much clearer. It should be noted that most of the existing methods make serious errors in the prediction of the red clutter part in Fig. 6(b). DenseASPP is able to

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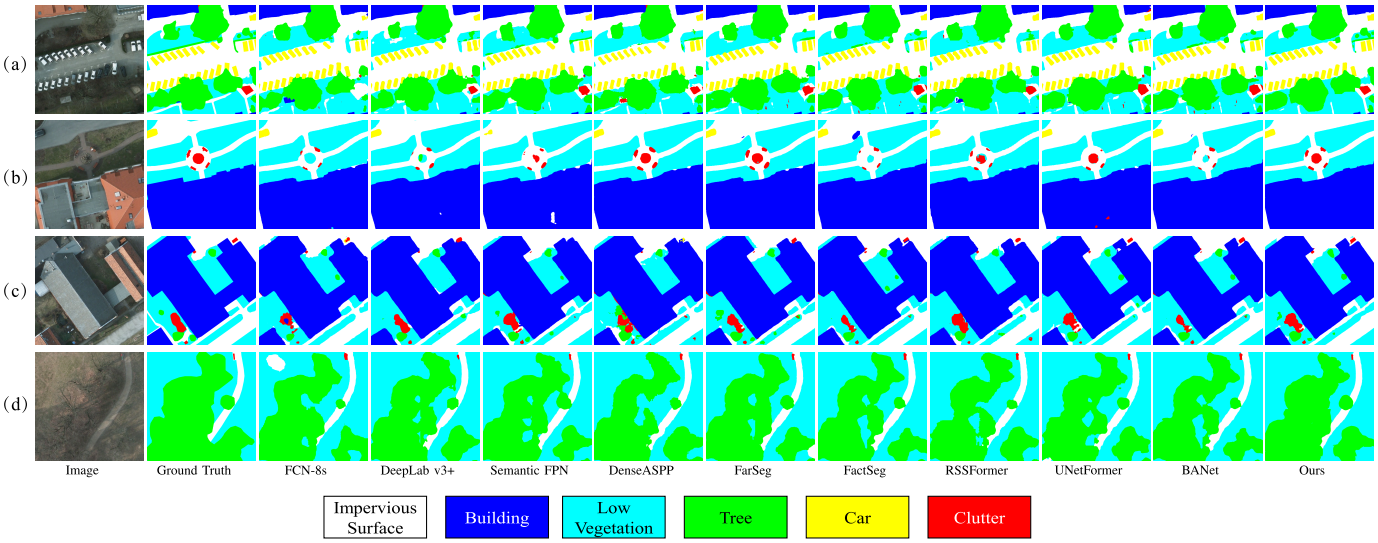


Fig. 6. Visualization results for the Potsdam validation set. (a) Car, tree, low vegetation, and building. (b) Car, low vegetation, building, and clutter. (c) Tree, low vegetation, building, and clutter. (d) Tree and low vegetation.

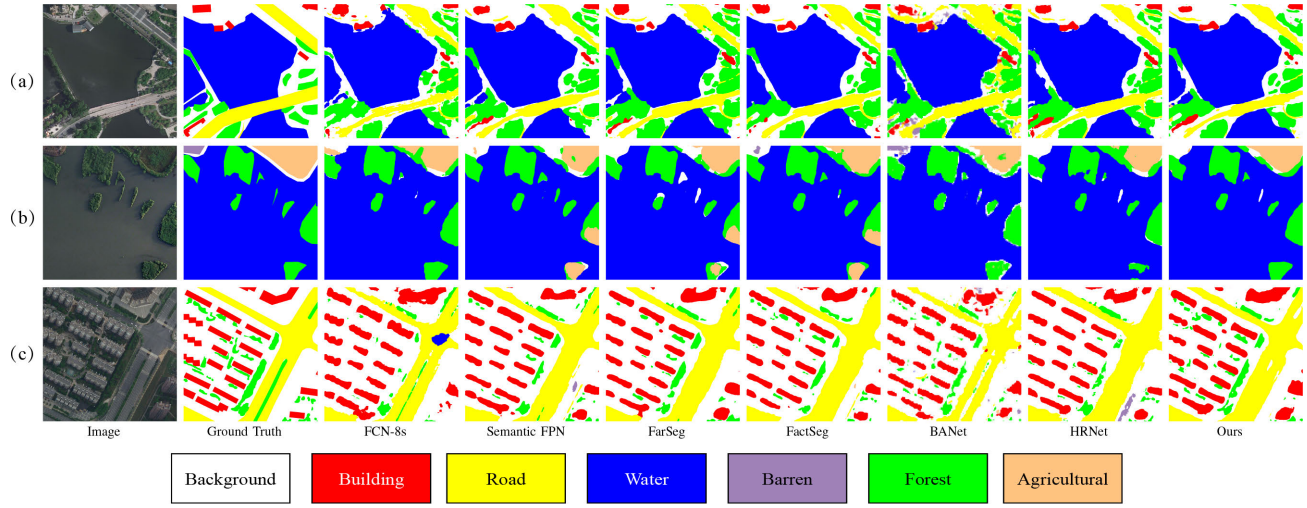


Fig. 7. Visualization results for the LoveDA validation set. (a) Building, road, water, and forest. (b) Water, barren, forest, and agricultural. (c) Building, road, and forest.

identify these objects completely, yet the generated boundary is very sloppy. This occurs because Potsdam suffers from a large intraclass variance. Moreover, the class clutter appears frequently in the whole dataset. Comparatively, our proposal is able to solve this problem exactly and even harvest the more accurate boundaries. Another challenge is the prediction of low vegetation in Fig. 6(d), which can be easily confused by the trees. The issue is often aggravated when the targets appear in close proximity. Again, only our method is able to provide relatively complete edges on low vegetation and trees. These experiments validate the effectiveness of our proposal.

3) Comparisons on the LoveDA Dataset: To explore the more possibilities of our method, here, we discuss the generalization of MCCANet. The experiments are conducted on LoveDA, which differs from the most semantic segmentation dataset, since it contains images from both urban and rural domains. The different geographical landscapes between the two make the distribution of the categories more inconsistent as well. By comparing MCCANet with FCN-8s [9], DeepLab V3+ [7], UNet [10], Semantic FPN [48], PSPNet [8], FarSeg [4], FactSeg [17], BANet [57], and HRNet [58], the qualitative and quantitative results are given in Fig. 7 and Table III, respectively. Encouragingly, our method again achieves the best results on both the validation and test sets. It can be intuitively found that, under the influence of the huge differences between urban and rural areas, including shape, size, and layout, the more serious FPs happen. As shown in Fig. 7(b), due to the severely intraclass variations, Semantic FPN and FactSeg misreport most of the forest component as agriculture. Accordingly, the results of our method still keep closely to the ground truth. Fig. 7(a) and (c) also illustrates that the MCCANet is the least misreported method for forest category, validating the superiority of our proposal.

D. Efficiency Analysis

The efficiency of a deep learning method can be evaluated by criteria, such as model params, giga floating-point operations per second (GFLOPs), and prediction time, which are also very important for the practical applications. We conduct a series of experiments to evaluate the efficiency of MCCANet.

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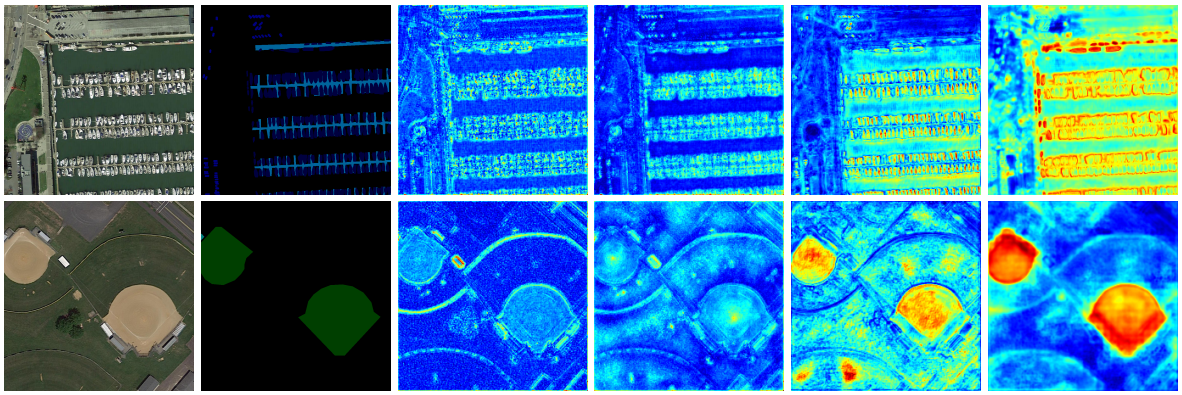


Fig. 8. Visualization of feature maps. (a) Original image. (b) GT. (c) Feature maps extracted by backbone. (d) Feature maps extracted by MCCA w/o CCS. (e) Feature maps extracted by MCCA w/ CCS. (f) Feature maps extracted by MCCANet.

TABLE III
COMPARISONS WITH STATE OF THE ARTS ON LOVEDA

Methods	Backbone	Val Set mIoU (%)	Test Set mIoU (%)
FCN-8s [9]	VGG-16	47.12	46.69
DeepLab V3+ [7]	ResNet-50	48.07	47.62
UNet [10]	ResNet-50	48.32	47.84
Semantic FPN [48]	ResNet-50	48.52	48.15
PSPNet [8]	ResNet-50	48.71	48.31
FarSeg [4]	ResNet-50	50.05	48.15
FactSeg [17]	ResNet-50	49.63	48.94
BANet [57]	ResT-Lite	49.62	49.63
HRNet [58]	W32	51.34	49.79
Ours	ResNet-50	51.87	50.36

For a single patch of size 896 × 896, the recorded params, GFLOPs, and prediction time are shown in Table IV. The experimental environment is with an NVIDIA RTX 3090 GPU and PyTorch platform. Note that the CCS subassembly, although impacts little on the overall prediction time, requires massive params for the computations of high-order magnitude of channels. Therefore, we report the results with and without consideration of CSS. Clearly, with CCS removed, our model occupies the first level in all aspects of efficiency. Comparatively, with CSS added, the efficiency of MCCANet still takes the middle place. Since the overall performance of our model has been proven to be state of the art, we believe this level of efficiency is acceptable.

E. Ablation Studies on the iSAID Dataset

1) Ablation for Module Verification: To evaluate the practical efficacy of different modules, i.e., MCCA, CCS, and BS, several ablation experiments are conducted in this section. Taking ResNet-50 + FPN as the baseline, the results of our model equipped with different modules are detailed in Table V. First, we replace the feature fusion block of FPN with MCCA. Then, the modules of CSS and BS are individually equipped to form the final MCCANet. It should be noted that the MCCANet selects multiple loss functions for model supervision, some of which need the facilitation of different modules. For the sake of fairness and uniformity, all experiments use only the CE loss for supervision. From Table V, one can see that the combination of MCCA and CCS brings a 3.6% improvement in terms of mIoU. Besides, the BS module can harvest a 1.5% improvement on top of the former. Finally, even using only CE loss, the use of all the modules results in a 64.8% mIoU.

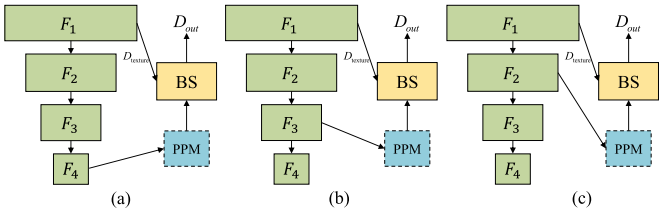


Fig. 9. Different BS strategies. (a)–(c) Use of BS modules on F_4, F_3, and F_2 stages, respectively. The dotted box indicates whether PPM is used.

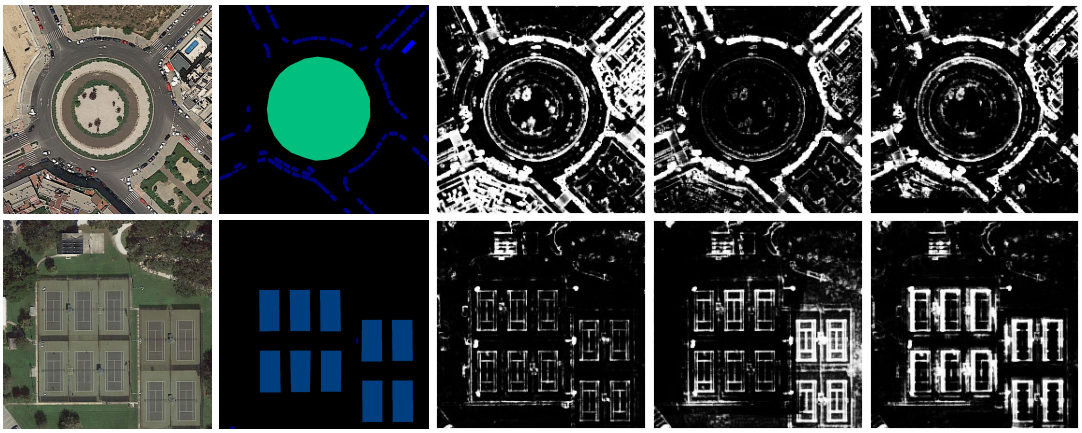
TABLE IV
COMPARISONS WITH STATE OF THE ARTS ON PARAMS, GFLOPS, AND PREDICTION TIME FOR A SINGLE PATCH

Method	Backbone	Params	GFLOPs	Time
FCN-8s [9]	VGG-16	50.52M	326	0.054s
DeepLab V3+ [7]	ResNet-50	39.17M	146	0.065s
ResUNet [56]	ResNet-50	51.04M	248	0.052s
Semantic FPN [48]	ResNet-50	28.47M	135	0.043s
PSPNet [8]	ResNet-50	53.31M	616	0.178s
FarSeg [4]	ResNet-50	31.37M	144	0.059s
FactSeg [17]	ResNet-50	33.44M	205	0.054s
DenseASPP [49]	Dense-121	9.18M	130	0.084s
HRNet [58]	W32	29.54M	138	0.075s
w/o CCS and BS	ResNet-50	26.99M	71	0.044s
MCCA w/o BS	ResNet-50	39.91M	190	0.056s
Ours	ResNet-50	42.49M	319	0.063s

Meanwhile, some visualized heat maps are illustrated in Fig. 8. Evidently, the backbone network is capable of extracting the general features of the given images, yet its capabilities in context modeling and semantic parsing are far from being available. With all playing the respective roles, the modules of MCCA, CCS, and BS assist in the network to better capture the target features and boundary information. The final heat map of the overall network clearly presents the more concrete foreground objects and highlights the saliency.

2) Ablation for Loss Functions: As known, loss functions play a key role during the model optimization [60], which

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(a) (b) (c) (d) (e)

Fig. 10. Visualization of boundary results under different supervision strategies. (a) Original image. (b) GT. (c) Supervision on F_2 stage. (d) Supervision on F_3 stage. (e) Supervision on F_4 stage.

TABLE V
ABLATION STUDY FOR MODULE VERIFICATION

Methods	MCCA	CCS	BS	mIoU(%)
Baseline	-	-	-	59.7
Ours	✓			61.5
	✓	✓		63.3
			✓	61.7
	✓	✓	✓	64.8

guides the network perform effective feature extraction. In the field of RS applications, the choice of loss function is more important due to the issues of large intraclass variations and extreme foreground–background imbalance. For instance, collaborative probability (CP) loss is tried to solve the unbalanced issue in FactSeg [17]. We also pay attention to the concerned issues and use $\mathcal{L}_{\text{seg}}$ to enable our network to focus more on small and hard-to-predict targets. Besides, the boundary recognition-related and category discrimination-related loss functions are further added to supervise the model optimization. To validate the necessity of different losses, we gradually add them to the CE baseline loss and conduct ablation experiments by fine-tuning the ratios among the hyperparameters. Table VI reports the specific results, where a hyperparameter value of 0 indicates that the corresponding loss is not considered. Overall, the incrementally added loss components lead to a significant improvement in performance. The last two columns of Table VI provide the evaluation results with the ratios of the three hyperparameters roughly adjusted. As can be seen, although the overall performance differences are mild, the best results come from the values being $\lambda_1 = 1$, $\lambda_2 = 2$, and $\lambda_3 = 1$, which is then considered as our first choice in practice.

3) Ablation Study for Different BS Strategies: The responsibility of BS is not only to model a better edge contour, but also to perform category supervision on the surrounding pixels. In general, the classification of pixel points around the boundary is much more challenging. Therefore, we try to explore the benefits of BS from ablation studies.

The feature maps of different stages of the backbone network often have different characteristics. While the low-level

TABLE VI
ABLATION STUDY FOR LOSS FUNCTIONS. $\mathcal{L}_{\text{CE}}$ DENOTES THE BASELINE WITH THE TRADITIONAL CE LOSS. THE THIRD AND FOURTH COLUMNS LIST THE RESULTS WITH ONLY "$\mathcal{L}_{\text{SEG}}$" AND "$\mathcal{L}_{\text{SEG}}$ PLUS $\mathcal{L}_{\text{BCE}}$," RESPECTIVELY

$\lambda_1 : \lambda_2 : \lambda_3$	$\mathcal{L}_{ce}$	1:0:0	1:0.5:0	1:2:0.5	1:1:1
			1:1:0	1:2:1	0.5:2:1
			1:2:0	1:2:2	2:2:1
			1:3:0	1:2:3	3:2:1
mIoU(%)	64.83	**65.74**	65.93	66.79	66.89
			65.98	**66.96**	66.69
			66.17	66.87	66.80
			66.05	66.62	66.77

TABLE VII
ABLATION STUDY FOR BS STRATEGIES

Methods	size(C×H×W)	w/o PPM mIoU(%)	w/ PPM mIoU(%)
Baseline	-	-	63.3
Stage F_2	512×112×112	64.9	66.4
Stage F_3	1024×56×56	65.3	66.6
Stage F_4	2048×28×28	65.8	66.9

features take charge of target localization, the high-level features are often responsible of target recognition. Therefore, different strategies of feeding the feature maps from backbone to the BS module deserve an in-depth investigation. As shown in Fig. 9, we consider the feeding of features from different stages, i.e., F_4, F_3, and F_2, to the BS module. Meanwhile, the role of PPM that aggregates the contextual information of different regions is also under consideration.

The visual results are shown in Fig. 10. As expected, low-level features have stronger localization ability. In the case of high resolution and small receptive field, the learned boundaries are in more detail, yet the discriminative power is greatly reduced due to the lack of semantic information. From Fig. 10(c), we can observe that the obtained boundary results from F_2 stage contain more useless details. The introduction of BS in the F_4 stage performs exactly the opposite result. With the stronger semantic information, the module is more aware

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of what we need to portray. Numerical results are presented in Table VII. Note the baseline refers to the model taking little consideration of the BS, yet with the MCCA module and its CCS subassembly equipped. On that basis, only the common CE loss could be imposed. Evidently, the role of PPM is positive, from which the learned global information favors the generated performance. Besides, the strategy imposed on the F_4 stage outcomes the best results.

V. Conclusion

For the semantic segmentation of HSR RS images, a new network architecture, namely, MCCANet, is proposed to alleviate the problems of extreme foreground–background imbalance and large intraclass variances. Overall, the framework adopts an effective feature fusion module (MCCA) to obtain multiscale contextual information and applies boundary and category information for supervision. Specifically, by varying the size of the spatial pool, MCCA enables CA to learn more scalewise information. Besides, with the aid of CCS module, the semantic gaps are eliminated as much as possible and achieve semantic pattern unification while handling intraclass difference. Finally, the BS module is applied to model object boundaries to recover the complete contours. The extensive experimental results on iSAID, Potsdam, and LoveDA datasets demonstrate the effectiveness and efficiency of our proposal. In the near future, we plan to simplify the MCCA module for an improved efficiency. Meanwhile, we try to explore the potential of MCCANet in more general applications, such as instance segmentation and change detection.

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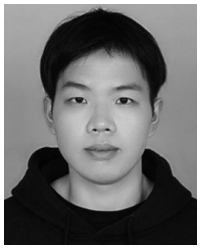
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